Lecture 5

Fusion categories: particles and how they fuse

Objectives. *Lectures 1–4 built the language: semisimple categories, tensor products, duals and loops. This lecture puts them together into the setting of anyon models. A fusion category has finitely many particle types. Two particles fuse into a sum of types, counted by multiplicities N_{ab}^{c}. States with a fixed total charge form* fusion spaces. *For repeated Fibonacci particles with total charge τ, their dimensions are Fibonacci numbers. We also count even numbers of Ising particles with vacuum charge.*

By the end of this lecture you will be able to:

- State the definition of a fusion category and give the fusion rules of Vec_G, $\mathrm{Rep}(S_3)$, Fibonacci, Ising and $\mathrm{SU}(2)_k$.
- Prove that the multiplicities form an associative fusion ring with $N_{ab}^{c} = N_{bc^*}^{a^*}$.
- Compute Frobenius–Perron and quantum dimensions and the global dimension.
- Use the trivalent vertex, the bubble and completeness relations, and count fusion spaces with fusion trees.

5.1 The definition

Definition 5.1 (Fusion category)**.** A *fusion category* is a $\mathbb{C}$-linear semisimple category with finitely many simple objects (Definition 2.11), with a monoidal structure whose tensor product is bilinear on morphisms, which is rigid (Definition 4.1), and whose unit 1 is simple. We take it pivotal (Definition 4.6), as all our examples are; then $X^{**} \cong X$.

The simple objects $a, b, c, \ldots$ are the *particle types*, or anyons; 1 is the vacuum, and the dual a^*, which we also write $\bar{a}$, is the antiparticle. The tensor product of two simple objects decomposes,

$$a \otimes b \cong \bigoplus_{c} N_{ab}^{c}\, c,$$

with *fusion multiplicities* $N_{ab}^{c} = \dim \mathrm{Hom}(a \otimes b, c)$ (Lemma 2.12).

Rigidity means that every particle can be created from the vacuum together with its antiparticle, by coev_a, and annihilated with it, by ev_a. $\mathrm{Rep}(\mathrm{SU}(2))$ has every property but finiteness, since its spins $j = 0, \frac{1}{2}, 1, \ldots$ do not stop. The category $\mathrm{Vect}_{\mathrm{fd}}$ is the trivial fusion category, whose only particle is the vacuum.

Example 5.2 (Fusion categories)**.**

- Vec_G^{ω} for a finite group G (Lecture 3): simple objects $g \in G$, $g \otimes h = gh$, and $g^* = g^{-1}$. Every simple object is invertible; such a category is *pointed*.

- $\mathrm{Rep}(G)$ for a finite group (Lectures 2 to 4): the fusion multiplicities are the Clebsch–Gordan multiplicities, computed from characters since $\chi_{V\otimes W} = \chi_V\chi_W$ (*Symmetries* Prop. 8.2), and V^* is the dual representation. For S_3, with simple objects $\mathbf{1}$, sgn, std, all self-dual (*Symmetries* Ex. 8.4):

$$\mathrm{sgn}\otimes\mathrm{sgn}\cong\mathbf{1},\quad \mathrm{sgn}\otimes\mathrm{std}\cong\mathrm{std},$$
$$\mathrm{std}\otimes\mathrm{std}\cong\mathbf{1}\oplus\mathrm{sgn}\oplus\mathrm{std}.$$

- *Fibonacci*: simple objects $1,\tau$, with

$$\tau\otimes\tau\cong 1\oplus\tau.$$

- *Ising*: simple objects $1,\sigma,\psi$, with

$$\sigma\otimes\sigma\cong 1\oplus\psi,$$
$$\sigma\otimes\psi\cong\sigma,\quad \psi\otimes\psi\cong 1.$$

- $\mathrm{SU}(2)_k$: simple objects $j=0,\frac{1}{2},\dots,\frac{k}{2}$, with the truncated Clebsch–Gordan rule

$$j_1\otimes j_2\cong\bigoplus_{j=|j_1-j_2|}^{\min(j_1+j_2,\,k-j_1-j_2)} j,$$

  in integer steps; without the truncation it is *Symmetries*' Thm. 14.12.

In the last three every simple object is self-dual: $N_{aa}^1=1$ for each a. For Fibonacci, Ising and $\mathrm{SU}(2)_k$ we have so far only the rules; that categories with these rules exist is shown in Lecture 6, by solving the pentagon for Fibonacci and verifying a solution for Ising, and for $\mathrm{SU}(2)_k$ in Lecture 9, from a cited construction.

5.2 Fusion rings

Theorem 5.3 (The fusion ring · *proved given coherence*)**.** *Let $K(C)$ be the free abelian group on the simple objects, with*

$$[a][b]=\sum_c N_{ab}^c\,[c].$$

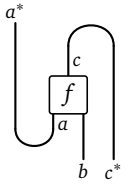
This product is associative with unit $[1]$, and

$$N_{ab}^1=\delta_{b,a^*},$$
$$N_{ab}^c=N_{bc^*}^{a^*},$$
$$N_{ab}^c=N_{b^*a^*}^{c^*}.$$

Proof. Associativity: $(a\otimes b)\otimes c\cong a\otimes(b\otimes c)$, and the multiplicity of e on the two sides is $\sum_x N_{ab}^x N_{xc}^e$ and $\sum_y N_{bc}^y N_{ay}^e$. The unit: $1\otimes a\cong a$. Bending wires (Theorem 4.5) gives $\mathrm{Hom}(a\otimes b,1)\cong\mathrm{Hom}(a,b^*)$, which by Schur is $\mathbb{C}$ if $a\cong b^*$ and 0 otherwise (b^* is simple, since the rotated box is a bijection $\mathrm{End}(b)\to\mathrm{End}(b^*)$); since $b^{**}\cong b$, $a\cong b^*$ exactly when $b\cong a^*$. For the second identity, bend c down and then a across, using its right dual (also a^*, since C is pivotal):

$$\mathrm{Hom}(a\otimes b,c)\cong\mathrm{Hom}(a\otimes b\otimes c^*,1)$$
$$\cong\mathrm{Hom}(b\otimes c^*,a^*),$$

and compare dimensions. In pictures, f goes to



whose inverse bends the two strands back. For the third, the rotated box gives $\mathrm{Hom}(a\otimes b, c) \cong \mathrm{Hom}(c^*, (a\otimes b)^*)$, and $(a\otimes b)^* \cong b^*\otimes a^*$ (Proposition 4.3); by semisimplicity the multiplicity of c^* in $b^*\otimes a^*$ is the dimension of either Hom space between them. ∎

For Fibonacci the two bracketings of three τ's give

$$(\tau\otimes\tau)\otimes\tau \cong (1\oplus\tau)\otimes\tau \cong 1\oplus 2\tau,$$
$$\tau\otimes(\tau\otimes\tau) \cong \tau\otimes(1\oplus\tau) \cong 1\oplus 2\tau.$$

The ring records that the two sides are isomorphic, but not which isomorphism the associator is; that is the content of Lecture 6.

The *fusion matrix* N_a is the matrix of left multiplication by $[a]$, $(N_a)_{cb} = N_{ab}^c$. For Fibonacci and Ising, in the bases $1, \tau$ and $1, \sigma, \psi$,

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad N_\sigma = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

and associativity reads $N_aN_b = \sum_c N_{ab}^c N_c$. The matrices need not commute: in Vec_G for $G = S_3$, $N_gN_h = N_{gh}$. The rings of our anyon models are commutative, because $a\otimes b \cong b\otimes a$ once there is a braiding (Lecture 7). A ring with a basis of this kind, non-negative integer structure constants and the properties of the theorem is a *fusion ring*.

Different categories can share a ring. For $k = 2$ the truncated rule gives $\frac{1}{2}\otimes\frac{1}{2} \cong 0\oplus 1$, $\frac{1}{2}\otimes 1 \cong \frac{1}{2}$ and $1\otimes 1 \cong 0$: under $0, \frac{1}{2}, 1 \mapsto 1, \sigma, \psi$ this is the Ising ring, though Lecture 6 shows that the two categories differ. For $k = 3$, $1\otimes 1 \cong 0\oplus 1$, since the truncation removes $j = 2$; the objects 0 and 1 fuse like 1 and τ.

The fusion ring does not determine the category. Vec_G and Vec_G^ω for $G = \mathbb{Z}_2$ have the same ring and are not equivalent (Lecture 3). For groups: the dihedral group of order 8 and the quaternion group have the same character table (*Symmetries* Ex. 5.15, which calls the dihedral group D_4), hence the same fusion ring, yet their representation categories are not equivalent as fusion categories (Calaque–Etingof, Cor. 3.12 · *cited*).

5.3 FPdim; the canonical spherical structure

The fusion matrices determine the Frobenius–Perron dimensions.

Theorem 5.4 (Frobenius–Perron dimension · *sketched given coherence*)**.** *For each simple a, the matrix N_a has non-negative entries and a largest real eigenvalue $\mathrm{FPdim}(a) \geq 1$ with a positive eigenvector. The map $a \mapsto \mathrm{FPdim}(a)$ is multiplicative,*

$$\mathrm{FPdim}(a)\,\mathrm{FPdim}(b) = \sum_c N_{ab}^c\,\mathrm{FPdim}(c),$$

and it is the only such function that is positive on every simple object (EGNO Prop. 3.3.6).

Sketch. Let $R = \sum_a [a]$, and let M be the matrix of right multiplication by R. Its entries $\sum_a N_{ba}^c = \sum_a \dim \mathrm{Hom}(a, b^* \otimes c)$ are positive, since $b^* \otimes c \neq 0$ (1 is a summand of $b \otimes b^*$, so c is one of $b \otimes b^* \otimes c$). By the Perron–Frobenius theorem M has a positive eigenvector v, unique up to scale, for its largest eigenvalue. Left and right multiplications commute, by associativity, so each N_a maps this eigenline to itself: $N_a v = \lambda_a v$, and $\lambda_a > 0$ because N_a is non-negative and non-zero. In fact $\lambda_a \geq 1$: every column of N_a has an entry at least 1, since $a \otimes b \neq 0$, so the entries of $N_a v$ add up to at least those of v. A non-negative matrix with a positive eigenvector has that eigenvalue as its largest, so $\lambda_a = \mathrm{FPdim}(a)$; applying $N_a N_b = \sum_c N_{ab}^c N_c$ to v gives multiplicativity. Conversely, a positive multiplicative f makes $(f(c^*))_c$ a positive eigenvector of M, so it is proportional to v; with $f(1) = 1$ this pins f down. ∎

For Fibonacci N_τ has eigenvalues $\varphi = (1+\sqrt{5})/2$ and $-1/\varphi$, so $\mathrm{FPdim}(\tau) = \varphi$; for Ising $\mathrm{FPdim}(\sigma) = \sqrt{2}$ and $\mathrm{FPdim}(\psi) = 1$; in $\mathrm{Rep}(G)$ it is the dimension. Uniqueness can be seen directly for Fibonacci: a multiplicative function has $d_\tau^2 = 1 + d_\tau$, whose only positive root is φ. The other root $-1/\varphi$ returns in Lecture 6 as the Yang–Lee category. For $\mathrm{SU}(2)_k$ (Sheet 5),

$$\mathrm{FPdim}(j) = \frac{\sin\big((2j+1)\pi/(k+2)\big)}{\sin\big(\pi/(k+2)\big)};$$

for $k = 2$ it gives $1, \sqrt{2}, 1$, Ising's values, and for $k = 3$ it gives $\mathrm{FPdim}(\frac{1}{2}) = \mathrm{FPdim}(1) = \varphi$.

The quantum dimension d_a of Lecture 4 is defined by a loop and depends on the pivotal structure. It too is multiplicative: $d_a d_b = d_{a\otimes b}$ (Theorem 4.7), and the trace is additive on the direct sum $a \otimes b \cong \bigoplus_c N_{ab}^c\, c$, so $d_a d_b = \sum_c N_{ab}^c d_c$. A fusion category is *pseudo-unitary* if $\sum_a d_a^2 = \sum_a \mathrm{FPdim}(a)^2$ for some spherical structure (EGNO Def. 9.4.4).

A fusion category is *unitary* if it has an antilinear dagger (§ 4.5) with $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$ and unitary associators and unitors. For every simple a and object X, require the formula

$$\langle v, w \rangle\ \mathrm{id}_a = v^\dagger w, \qquad v, w \in \mathrm{Hom}(a, X),$$

to define a positive inner product. Unitary fusion categories are pseudo-unitary (EGNO Rem. 9.4.7 · *cited*).

> **Theorem 5.5** (The canonical spherical structure · *cited*). *A pseudo-unitary fusion category has a unique spherical structure with $d_a = \mathrm{FPdim}(a)$ for every simple a (EGNO Prop. 9.5.1).*

All the anyon numerics of the course use this canonical structure (so $d_\tau = \varphi$, $d_\sigma = \sqrt{2}$). Other pivotal structures exist: on Vec_G for $G = \mathbb{Z}_2$ the choice $a_s = -\mathrm{id}$ of Lecture 4 gives $d_s = -1$. The *global dimension* is

$$\dim C = \sum_a d_a^2,$$

with the canonical d: $|G|$ for Vec_G and for $\mathrm{Rep}(G)$ (*Symmetries* Thm. 5.11), $2+\varphi$ for Fibonacci, 4 for Ising.

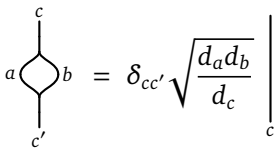
5.4 Vertices, bubbles, completeness

From here on the categories are unitary, with the canonical spherical structure (so every $d_a > 0$), and *multiplicity-free*, every $N_{ab}^c \in \{0, 1\}$, as all the course's anyon models are. Then

$\mathrm{Hom}(c, a \otimes b)$ is a line when $N_{ab}^{c} = 1$, and we choose a basis vector, the *splitting vertex* ψ_c^{ab}, drawn with c below; its dagger, the *fusion vertex* $\psi_c^{ab\dagger}$, is the picture reflected top to bottom:



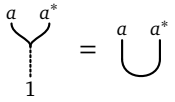
The vertex is fixed up to a scalar, and we fix its size by the *isotopy-invariant normalisation* (Bonderson (2.14)–(2.19); Kitaev eqs. 195–196): a fusion vertex on a splitting vertex is



(5.1)

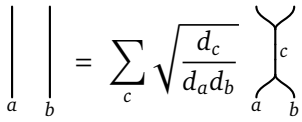
This can be arranged: by Schur the left side is a multiple of id_c, zero unless $c = c'$, and positive since it has the form $\psi^\dagger\psi$; rescaling ψ by a positive number sets it. A phase remains free, the gauge of Lecture 6.

The vacuum channels are fixed by Lecture 4. Choosing $\psi_a^{1a} = \psi_a^{a1} = \mathrm{id}_a$ satisfies (5.1). For $a \otimes a^* \to 1$, the bubble asks $\psi^\dagger\psi = \sqrt{d_a d_{a^*}} = d_a$. Choose the duality maps *standard*: each cap is the dagger of a cup, $\mathrm{coev}_a^\dagger = \widetilde{\mathrm{ev}}_a$, as in $\mathrm{Hilb}_{\mathrm{fd}}$ (Theorem 4.10); this can be done with the canonical dimensions (Kitaev eqs. (190)–(191) · *cited*). Then $\mathrm{coev}_a^\dagger\,\mathrm{coev}_a$ is the loop d_a, and we may take the cup itself:



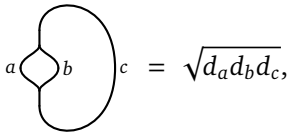
(5.2)

For a self-dual a this identifies a^* with a; when $\kappa_a = -1$, as for spin $\frac{1}{2}$ in $\mathrm{SU}(2)_k$, the cup carries the flag of Lecture 4. Since $a \otimes b$ is the direct sum of its channels, the identity of $a \otimes b$ is the sum of the projections onto them:



(5.3)

the coefficient making each term an idempotent by (5.1). Closing the c strand of (5.1) into a loop gives the *theta graph*,



(5.4)

symmetric in a, b, c. Taking the trace of (5.3), the left side becomes two loops, $d_a d_b$; by cyclicity of the trace each term on the right becomes $\sqrt{d_c/d_a d_b}$ times a theta graph, that is d_c. So

$$d_a d_b = \sum_c N_{ab}^{c}\, d_c,$$

recovering the multiplicativity of § 5.3.

In $\mathrm{Rep}(S_3)$, for std ⊗ std, a splitting vertex is an isometric intertwiner $c \to \mathrm{std} \otimes \mathrm{std}$ times $(4/d_c)^{1/4}$, and both relations can be checked as matrix identities. With std real orthogonal on an orthonormal basis e_1, e_2, the vacuum vertex is $1 \mapsto e_1 \otimes e_1 + e_2 \otimes e_2$, which is $\mathrm{coev}_{\mathrm{std}}$ under the intertwiner $\mathrm{std}^* \to \mathrm{std}$, $e^i \mapsto e_i$, and its bubble is $2 = d_{\mathrm{std}}$. Kitaev's App. E.1 uses unit-norm vertices instead, $\psi^\dagger\psi = \mathrm{id}$, with no square roots in completeness; the two conventions differ by $(d_c/d_a d_b)^{1/4}$ per vertex, and a formula must not mix them.

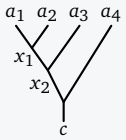
5.5 Fusion spaces

The states of n anyons $a_1, \dots, a_n$ with total charge c form the *fusion space*

$$V_c^{a_1 \cdots a_n} = \mathrm{Hom}(c, a_1 \otimes \cdots \otimes a_n).$$

For two spin-$\frac{1}{2}$ particles of total spin j this is the multiplicity space of the addition of angular momenta (*QM-I* Thm. 13.2).

Theorem 5.6 (Fusion trees). *Fix the left-associated bracketing. The* fusion trees *of this shape with root c and leaves $a_1, \dots, a_n$, labelled by intermediate simple objects $x_1, \dots, x_{n-2}$ with every vertex allowed, form a basis of $V_c^{a_1 \cdots a_n}$:*



So

$$\dim V_c^{a_1 \cdots a_n} = \sum_x N_{a_1 a_2}^{x_1} N_{x_1 a_3}^{x_2} \cdots N_{x_{n-2} a_n}^{c}.$$

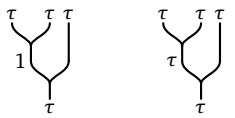
Proof. For $n = 2$ the space is a line or zero. For $n > 2$, decompose $a_1 \otimes \cdots \otimes a_{n-1}$ into simple objects by induction: by semisimplicity,

$$\begin{aligned} &\mathrm{Hom}(c, a_1 \otimes \cdots \otimes a_n) \\ &\quad \cong \bigoplus_x \mathrm{Hom}(c, x \otimes a_n) \\ &\qquad \otimes \mathrm{Hom}(x, a_1 \otimes \cdots \otimes a_{n-1}), \end{aligned}$$

and a basis is the vertex $\psi_c^{x a_n}$ followed by the trees of the second factor. ∎

Bracketing the leaves differently gives another basis of the same space; the change of basis is the F-move of Lecture 6.

Three τ's with total charge τ have two trees, with $x_1 = 1$ and $x_1 = \tau$:



In general each intermediate label of n τ's is 1 or τ, and a 1 cannot follow a 1. Write f_n for $\dim V_\tau^{\tau \cdots \tau}$ with n leaves; the last vertex is $\tau \otimes \tau \to \tau$ or $1 \otimes \tau \to \tau$, and a 1 after $n-1$ leaves must come from τ after $n-2$, so

$$f_n = f_{n-1} + f_{n-2}, \qquad f_1 = f_2 = 1.$$

The dimensions are $1, 1, 2, 3, 5, 8, \dots$, the Fibonacci numbers, and they grow like φ^n (Sheet 5): a large degenerate space, protected by topology. For Ising, $2m$ copies of σ with total charge 1 have 2^{m-1} states. For $m = 2$ the trees have $x_1 \in \{1, \psi\}$ and then $x_2 = \sigma$: four σ's carry one qubit.

Summary.

- *Handed on:* a fusion category is semisimple, rigid and pivotal, with finitely many simples, a simple unit and bilinear $\otimes$ (Definition 5.1); Fibonacci, Ising, $\mathrm{SU}(2)_k$, Vec_G^ω and $\mathrm{Rep}(G)$ are the examples.
- *The fusion ring:* N_{ab}^c form an associative ring with $N_{ab}^1 = \delta_{b,a^*}$ and $N_{ab}^c = N_{bc^*}^{a^*}$ (Theorem 5.3); it does not determine the category.
- *Dimensions:* FPdim is the Perron–Frobenius eigenvalue, φ for τ and $\sqrt{2}$ for σ; for a pseudo-unitary C with its canonical spherical structure $d = \mathrm{FPdim}$, and $\dim C = \sum_a d_a^2$ (Theorems 5.4 and 5.5).
- *Pictures:* the trivalent vertex with the bubble $\sqrt{d_a d_b / d_c}$ and completeness (5.1), (5.3); admissibly labelled trees of one fixed shape form a basis, with Fibonacci numbers for τ (Theorem 5.6).
- *Next:* Lecture 6 computes the associator on fusion trees, the F-symbols, and solves the pentagon for Fibonacci.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: fusion categories, Def. 4.1.1, p. 65; based and fusion rings, Defs. 3.1.3, 3.1.7 and Prop. 3.1.6, pp. 49–50; the Frobenius–Perron dimension, Def. 3.3.3 and Prop. 3.3.6, p. 53; the Yang–Lee (Fibonacci), Ising and Verlinde rings, Exs. 3.1.9 and 4.10.6, pp. 51 and 79; pseudo-unitary categories, Def. 9.4.4, p. 283, and unitary ones, Rem. 9.4.7, pp. 283–284; the canonical spherical structure, Prop. 9.5.1, p. 284. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438, App. E: fusion rules, p. 67, unit-norm vertices, eqs. 174–176, p. 68, the standard cups and caps, eqs. (190)–(191), p. 76, the isotopy-invariant normalisation, eqs. 195–196, p. 77, quantum dimensions as Perron–Frobenius eigenvalues, pp. 79–80. Bonderson, PhD thesis (Caltech, 2007), (2.1)–(2.19), pp. 13–19. Simon, *Topological Quantum*, Ch. 8, pp. 93–102 (Fibonacci counting, p. 98). Calaque and Etingof, *Lectures on tensor categories*, Cor. 3.12, p. 24.